

Assignment 2

Indian Institute of Information Technology Pune
Machine Learning Lab

Q1.

```
import numpy as np
```

```
A = np.array([ [12, 5, 8, 20], [ 7, 15, 3, 11], [25, 9, 18, 6], [10, 14, 2, 16] ])
```

Perform the following without using explicit Python for loops:

1. Calculate the mean of each row.
2. Replace every element that is greater than its row mean by 1.
3. Replace every other element by 0.
4. Return the resulting binary matrix.
5. Also determine which row contains the maximum number of elements above its row mean.

Q2.

```
X = np.array([ [10, 100, 5], [12, 110, 6], [11, 105, 7], [13, 500, 6], [14, 108, 8], [15, 115, 9] ])
```

For each column:

1. Calculate the mean and standard deviation.
2. Identify values satisfying:

$$|x - \mu| > 2\sigma$$

3. Replace such values with the corresponding column median.
4. Do this without modifying non-outlier values.

Q3.

```
import pandas as pd
```

```
df = pd.DataFrame({ 'Student': ['A','B','C','D','E','F','G','H'],  
  'Department': ['CSE','CSE','ECE','ECE','CSE','ECE','CSE','ECE'],  
  'Gender': ['F','M','F','M','M','F','F','M'],  
  'Marks': [85,72,91,65,78,88,95,70],  
  'Attendance': [92,81,96,72,85,90,98,75] })
```

Find:

1. Average marks for each department.
2. Average marks separately for each (Department , Gender) group.
3. Student with the highest marks in each department.
4. Students whose marks are above their department average.
5. Department having the highest average attendance.

Q4.

```
df = pd.DataFrame({ 'ID': [101,102,103,101,104,102,105],  
  'Name': ['A','B','C','A','D','B','E'],  
  'Age': [20,21,22,20,23,21,24],  
  'Score': [80,75,90,85,88,75,92] })
```

Perform:

1. Identify duplicated IDs.
2. For each ID, determine whether its records are identical.
3. If multiple records exist for an ID, retain the record having the highest Score.
4. Produce a clean DataFrame containing exactly one record per ID.
5. Report how many records were removed.

Q5.

```
df = pd.DataFrame({ 'Date': [ '2026-01-01',  
  '2026-01-02',  
  '2026-01-04',  
  '2026-01-05',  
  '2026-01-08',  
  '2026-01-09' ],  
  'Sales': [100,120,150,130,180,200] })
```

Perform:

1. Convert Date into a proper datetime column.
2. Detect the missing dates.
3. Reindex the DataFrame to contain every date between the minimum and maximum date.

4. Keep missing sales as NaN.
5. Calculate a 3-day rolling average of sales.
6. Identify the date on which the rolling average is maximum.

Q6.

```
P = np.array([ [1, 2], [4, 6], [7, 3], [2, 8], [9, 5] ])
```

Construct a 5×5 matrix where element (*i* , *j*) contains the Euclidean distance between points *i* and *j* .

The solution must:

- use NumPy broadcasting;
- contain no for or while loop;
- have zeros on the diagonal;
- produce a symmetric matrix.

Then identify the pair of points having the maximum distance.

Q7. Take 10 datasets from kaggle (Tabular datasets e.g. weather dataset, housing dataset, titanic dataset, etc.) Perform exploratory data analysis on the dataset using seaborn , matplotlib. (Draw graphs like bar chart, line chart, scatter plots, etc. and explain the insights). Also clean the dataset accordingly.